Population Genetics Simulator Pro

Upload a CSV of parameters — simulate, visualize, and analyze in bulk | Wright-Fisher neutral model

Simulation Results

500ind_10Kgen_0p5AF

A_1 Lost

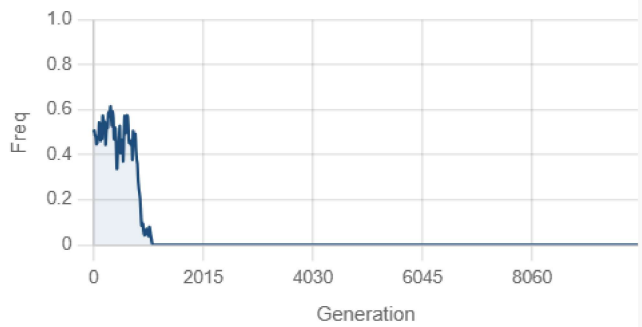
N = 500 | 10,000 gen | p_0 = 0.5

FINAL A_1 FREQ	AVG FREQ	SD (DRIFT)
0.0000	0.0424	0.1356

FIX / LOSS

Gen 1,086

ALLELE FREQUENCY (A_1)



10Kind_10Kgen_0p5AF

Polymorphic

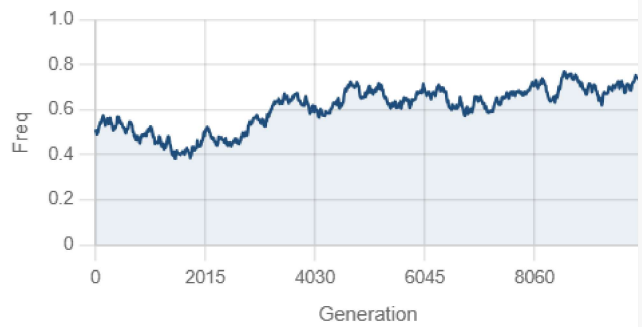
N = 10,000 | 10,000 gen | p_0 = 0.5

FINAL A_1 FREQ	AVG FREQ	SD (DRIFT)
0.7379	0.6063	0.0942

FIX / LOSS

—

ALLELE FREQUENCY (A_1)



Infinind_10Kgen_0p5AF

Stable (∞)

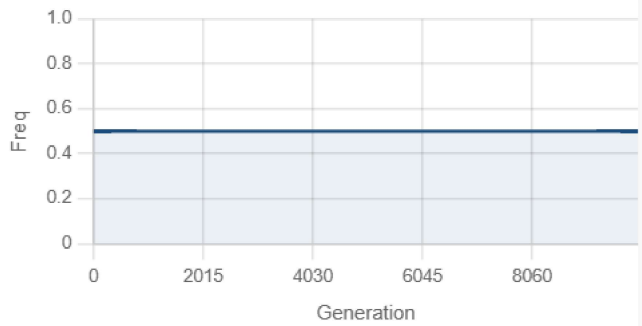
N = ∞ | 10,000 gen | p_0 = 0.5

FINAL A_1 FREQ	AVG FREQ	SD (DRIFT)
0.5000	0.5000	0.0000

FIX / LOSS

—

ALLELE FREQUENCY (A_1)



500ind_10Kgen_0p1AF

A_1 Lost

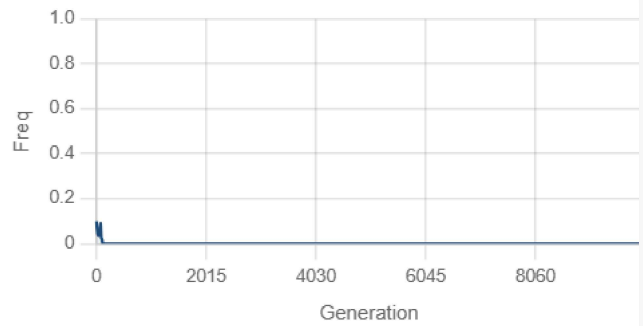
N = 500 | 10,000 gen | p_0 = 0.1

FINAL A_1 FREQ	AVG FREQ	SD (DRIFT)
0.0000	0.0006	0.0060

FIX / LOSS

Gen 111

ALLELE FREQUENCY (A_1)



10Kind_10Kgen_0p1AF

Polymorphic

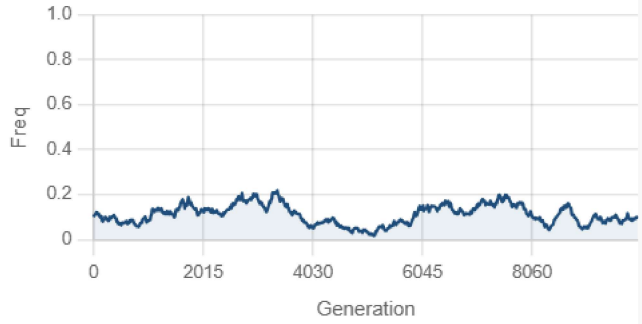
N = 10,000 | 10,000 gen | p_0 = 0.1

FINAL A_1 FREQ	AVG FREQ	SD (DRIFT)
0.0936	0.1117	0.0432

FIX / LOSS

—

ALLELE FREQUENCY (A_1)



Infinind_10Kgen_0p1AF

Stable (∞)

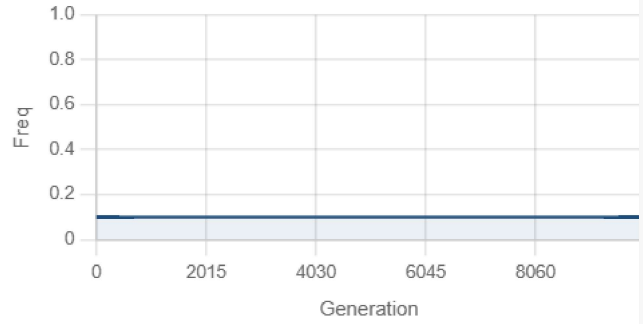
N = ∞ | 10,000 gen | p_0 = 0.1

FINAL A_1 FREQ	AVG FREQ	SD (DRIFT)
0.1000	0.1000	0.0000

FIX / LOSS

—

ALLELE FREQUENCY (A_1)



500ind_10Kgen_0p9AF

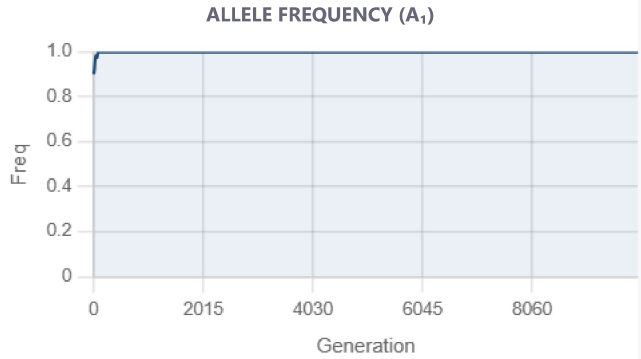
A_1 Fixed

N = 500 | 10,000 gen | p_0 = 0.9

FINAL A_1 FREQ	AVG FREQ	SD (DRIFT)
1.0000	0.9997	0.0045

FIX / LOSS

Gen 86



10Kind_10Kgen_0p9AF

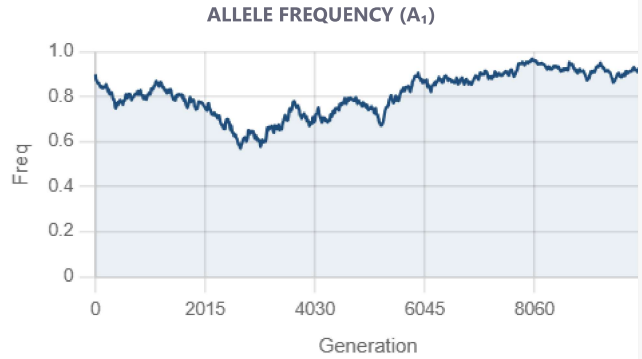
Polymorphic

N = 10,000 | 10,000 gen | p_0 = 0.9

FINAL A_1 FREQ	AVG FREQ	SD (DRIFT)
0.9204	0.8130	0.0967

FIX / LOSS

—



Infinind_10Kgen_0p9AF

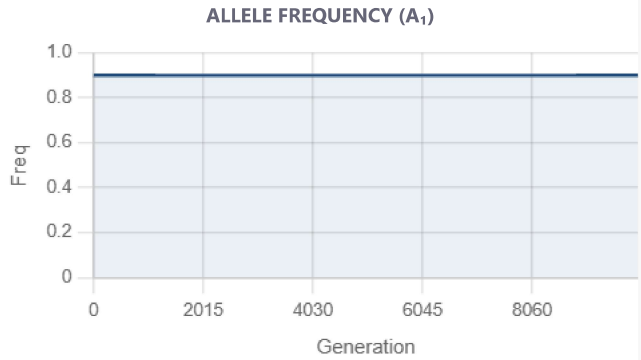
Stable (∞)

N = ∞ | 10,000 gen | p_0 = 0.9

FINAL A_1 FREQ	AVG FREQ	SD (DRIFT)
0.9000	0.9000	0.0000

FIX / LOSS

—



Summary Statistics

Simulation	N	Generations	p_0	Final Freq	Avg Freq	SD	Outcome
500ind_10Kgen_0p5AF	500	10,000	0.5	0.0000	0.0424	0.1356	A_1 Lost

Simulation	N	Generations	p_0	Final Freq	Avg Freq	SD	Outcome
10Kind_10Kgen_0p5AF	10,000	10,000	0.5	0.7379	0.6063	0.0942	**Polymorphic**
Infinind_10Kgen_0p5AF	∞	10,000	0.5	0.5000	0.5000	0.0000	**Stable (∞)**
500ind_10Kgen_0p1AF	500	10,000	0.1	0.0000	0.0006	0.0060	**A_1 Lost**
10Kind_10Kgen_0p1AF	10,000	10,000	0.1	0.0936	0.1117	0.0432	**Polymorphic**
Infinind_10Kgen_0p1AF	∞	10,000	0.1	0.1000	0.1000	0.0000	**Stable (∞)**
500ind_10Kgen_0p9AF	500	10,000	0.9	1.0000	0.9997	0.0045	**A_1 Fixed**
10Kind_10Kgen_0p9AF	10,000	10,000	0.9	0.9204	0.8130	0.0967	**Polymorphic**

Simulation Analysis

Auto-generated from simulation outcomes. All runs under neutral conditions (no selection, mutation, migration, inbreeding, or assortative mating).

Overview

9 simulations were run under neutral conditions (no selection, mutation, migration, inbreeding, or assortative mating). Starting A_1 frequencies tested: p_0 = 0.500, p_0 = 0.100, p_0 = 0.900. Population sizes: 500, 10,000, ∞. Duration: 10,000 generations.

Effect of Population Size on Genetic Drift

Genetic drift is the random fluctuation of allele frequencies caused by chance variation in which individuals reproduce. Its strength depends directly on population size through the per-generation sampling variance:

```
Var(Δp) = p(1 - p) / 2N
```

As N decreases, the variance increases, producing larger random swings in allele frequency each generation and accelerating fixation or loss.

> **At p_0 = 0.500:**
> - **N = 500:** A_1 **lost** at generation 1,086 (SD = 0.1356) — rapid drift to loss
> - **N = 10,000:** Remained **polymorphic** through generation 10,000 (final freq = 0.738, SD = 0.0942) — drift too weak to cause fixation

> **At p_0 = 0.100:**
> - **N = 500:** A_1 **lost** at generation 111 (SD = 0.0060) — rapid drift to loss

- **N = 10,000:** Remained **polymorphic** through generation 10,000 (final freq = 0.094, SD = 0.0432) — drift too weak to cause fixation

At p_0 = 0.900:

- **N = 500:** A_1 **fixed** at generation 86 (SD = 0.0045) — rapid drift to fixation
- **N = 10,000:** Remained **polymorphic** through generation 10,000 (final freq = 0.920, SD = 0.0967) — drift too weak to cause fixation

The frequency standard deviation (SD) directly quantifies drift intensity. At N = 500, average SD across simulations = **0.0487**. At N = 10,000, average SD across simulations = **0.0780**. Smaller populations consistently show higher SD, confirming stronger drift.

Infinite Population: Hardy-Weinberg Equilibrium

When N = ∞, the sampling variance per generation approaches zero:

```
Var(Δp) = p(1 − p) / 2N → 0 as N → ∞
```

With no sampling error, allele frequencies remain exactly constant — the defining condition of Hardy-Weinberg equilibrium. Genotype frequencies stay fixed at p^2 (A_1A_1), 2pq (A_1A_2), and q^2 (A_2A_2) throughout all generations.

Infinite population results:

- **Infinind_10Kgen_0p5AF:** A_1 held exactly at p = 0.5 for all 10,000 generations (SD = 0.000000). Genotypes: A_1A_1 = 0.2500, A_1A_2 = 0.5000, A_2A_2 = 0.2500.
- **Infinind_10Kgen_0p1AF:** A_1 held exactly at p = 0.1 for all 10,000 generations (SD = 0.000000). Genotypes: A_1A_1 = 0.0100, A_1A_2 = 0.1800, A_2A_2 = 0.8100.
- **Infinind_10Kgen_0p9AF:** A_1 held exactly at p = 0.9 for all 10,000 generations (SD = 0.000000). Genotypes: A_1A_1 = 0.8100, A_1A_2 = 0.1800, A_2A_2 = 0.0100.

Role of Starting Allele Frequency

When drift is minimal and no evolutionary forces act, the **starting allele frequency (p_0)** is the primary determinant of the average final frequency. Drift is an unbiased process: the expected final frequency across many replicate populations always equals p_0, and the probability of fixation under pure drift = p_0.

Average final frequency by starting frequency:

- **p_0 = 0.500:** avg final freq = 0.4126 | 0 fixed, 1 lost, 2 polymorphic/stable
- **p_0 = 0.100:** avg final freq = 0.0645 | 0 fixed, 1 lost, 2 polymorphic/stable
- **p_0 = 0.900:** avg final freq = 0.9401 | 1 fixed, 0 lost, 2 polymorphic/stable

Higher starting frequencies yield higher average final frequencies, consistent with the neutral expectation that p_0 predicts long-run outcomes.

Conclusions

1. **Population size governs drift strength.** Smaller populations experience larger per-generation sampling variance [$Var = p(1-p)/2N$], leading to rapid and dramatic allele frequency changes. Larger populations buffer against drift, keeping allele frequencies near their starting values for far longer.
2. **Infinite populations maintain Hardy-Weinberg equilibrium.** Genetic drift requires finite sampling to operate. At $N = \infty$, allele and genotype frequencies remain perfectly stable indefinitely, satisfying HWE exactly.
3. **Starting allele frequency determines average long-run outcomes.** In the absence of selection and with drift minimized, p_0 sets the expected final frequency and equals the probability of fixation, because drift is directionally neutral on average.